Literature Review

Real-Time IoT Data Pipelines in Cold Chain Fleet Management

1. Introduction & Contextual Background

In modern logistics and supply chain management, the transport of temperature-sensitive goods—such as pharmaceuticals, perishable foods, and bio-chemical products—requires strict environmental control, known as the Cold Chain. Historically, fleet management relied on static logging devices where data was retrieved post-delivery. However, with the rise of the Internet of Things (IoT) and modern Data Engineering practices, logistics operations have shifted toward continuous, real-time telemetry to prevent inventory spoilage, minimize delays, and optimize route efficiency.

2. Evolution of Telematics: From Batch Processing to Streaming Pipelines

Traditional telematics systems utilized Batch **ETL** (Extract, Transform, Load) architectures. Vehicle location data and sensor readings were uploaded periodically (hourly or at the end of a shift) to relational databases.

Recent literature highlights the limitations of batch architectures in time-critical domains:

- **High Latency:** Critical failures, such as refrigeration breakdown or route deviation, are detected too late to take corrective action.
- **Scalability Bottlenecks:** Standard relational databases struggle with high-frequency write loads generated by thousands of concurrent IoT sensors.

To address these challenges, contemporary research emphasizes **Event-Driven Architecture (EDA)** and **Stream Processing**, where data is treated as an unbounded stream of events ingested and processed in near real-time.

3. Identification of Technical & Operational Gaps (The Research Gap)

By reviewing existing open-source and commercial fleet tracking solutions, three major technical gaps were identified:

1. **Isolated Data Silos:** Most existing systems separate GPS location tracking from operational metrics (maintenance costs, weather conditions, customer ratings), creating fragmented views for decision-makers.

2. **Suboptimal Time-Series Storage:** Conventional database schemas are not optimized for write-heavy IoT workloads, leading to index degradation and slow query response times over large historical datasets.
3. **Lack of Unified Anomaly Detection:** Systems often alert users after an incident occurs rather than evaluating rolling averages or real-time thresholds (temperature threshold violations during transit).

4. Architectural Justification & Methodology Selection

Our project design addresses these identified gaps by selecting lightweight, highly scalable, and time-series-optimized technologies:

- **Protocol Selection (MQTT vs. HTTP):** Literature consistently proves that **MQTT (Message Queuing Telemetry Transport)** is superior for IoT telemetry due to its minimal packet overhead, publish-subscribe model, and low bandwidth consumption compared to RESTful HTTP APIs.
- **Storage Optimization (TimescaleDB):** Standard relational databases struggle with real-time time-series data. **TimescaleDB** (a PostgreSQL extension) introduces *Hypertables*, which automatically partition time-series data across time intervals and space, enabling high-throughput writes and rapid time-bucket aggregation.
- **Visualization (Grafana):** Grafana is selected over traditional BI tools due to its native support for real-time streaming queries, interactive geospatial (map) dashboards, and built-in threshold alerting mechanisms.

5. Conceptual Framework & Key Evaluation Metrics

The conceptual framework for our pipeline is evaluated against three core data engineering metrics:

1. **End-to-End Latency:** The time elapsed from when an IoT sensor generates a telemetry payload to when it is stored and displayed on the dashboard.
2. **Ingestion Throughput:** The pipeline's capacity to handle concurrent streams of vehicle telemetry without message loss or queue backpressure.
3. **Data Completeness & Integrity:** Ensuring raw messages from the MQTT broker are cleaned, formatted, and correctly mapped across normalized database entities.

6. Conclusion

This Literature Review establishes that moving from batch-oriented telematics to a unified, real-time IoT pipeline is essential for modern cold chain logistics. By synthesizing event-driven ingestion (MQTT), time-series storage optimization (TimescaleDB), and real-time visualization (Grafana), our project directly bridges the gap between raw vehicle telemetry and actionable operational insights.